

The Effect of Narrative on Empathy: A Systematic Review and Meta-Analysis

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Abstract

Narratives are comprehended and processed differently than non-narrative messages. They require the reader to imagine the events of the narrative from the perspective of the story characters in the time and place where they occurred. Adopting this imagine-other perspective might elicit empathic concern. This systematic review and meta-analysis examined the effect of narrative on empathic concern. Eligible studies were experimental comparisons of narrative and non-narrative texts with similar content that measured empathic concern post-intervention. A meta-analysis of 5 eligible studies ($n = 1,406$) revealed a significant difference in empathic concern between narrative and non-narrative conditions ($d = 0.56, p = .036$). This effect was moderated by measurement instrument; the effect of narrative was large among studies using Batson et al.'s (1997) measure of state empathic concern ($d = 0.90$) and near zero among studies using the Interpersonal Reactivity Index (Davis, 1983), which measures dispositional empathic concern ($d = 0.02$). These findings suggest that narrative messages cause readers to experience greater empathy but reading a single narrative does not change readers' propensities for experiencing empathic concern. Only five eligible studies were identified for this review, suggesting this area is in need of further replications. In addition to replicating the present findings, future studies should seek to identify the specific narrative elements necessary for eliciting empathic concern.

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When, as a young child, Fred Rogers was scared by something he saw on the news, his mother would tell him to, “look for the helpers. You will always find people who are helping.” Reflecting on this advice in adulthood, Rogers said, “to this day, especially in times of disaster, I remember my mother’s words, and I am always comforted by realizing that there are still so many helpers—so many caring people in this world” (Family Communications, Inc, 2005). Rogers’ mother was right to think that there are always people helping, especially in the aftermath of disasters like those Rogers would have witnessed in the news (Kaniasty et al., 2020). But not everyone gets help in their times of need. Why is it that some people or causes receive outpourings of social support, and others do not?

Prior research suggests that altruistic helping happens when a person perceives another person as in need and values that person’s welfare (Batson, 2023a). When these conditions are met, the observer will experience empathic concern—an other-oriented emotion state characterized by feelings of sympathy and compassion—and want to help for no other reason than to make life better for the person they see as in need. This is the basic tenet of the empathy–altruism hypothesis: people help those with whom they empathize (Batson, 2023b; Batson et al., 1991).

The empathy–altruism hypothesis has a strong empirical basis. Experimental and observational studies find a small, positive association between feelings of empathic concern and helping (for reviews, see Batson, 2010; Eisenberg & Miller, 1987). This association suggests that empathic concern causes helping, it does not necessarily mean

that this helping is altruistic. Behaving altruistically means the reason for helping is to benefit someone else. Altruistic actions can still be personally beneficial, but personal benefit cannot be the reason for helping. People feeling empathic concern might have self-interested motivations for helping, such as influencing how one appears to others (impression management) and sharing in the joy of another's success (empathic joy); however, empathy still causes helping in experimental studies in which the possibility of these and other personal benefits are taken away (Batson, 2010, 2023b). As such, although there may be a selfish reason behind the helping that follows empathy, the current research suggests such help stems from a selfless desire to benefit others.

If empathic concern causes altruistic helping, it is important to understand what causes empathic concern. At minimum this requires observing an individual who appears to be in need (Batson, 2023c); however, conveying information about a person in need may be done in many different ways. In written accounts of individuals in need, narratives may be more effective than non-narratives at eliciting empathic concern. Although both kinds of texts can convey that an individual or group is in need, narratives may do so in a way that inspires readers to imaginatively simulate narrative events such that they feel close and real (Gerrig, 1998; Lee, 2004). As such, eliciting empathic concern (and by extension altruistic helping) may sometimes depend on telling a good story.

Narratives

What are narratives and how are they distinct from other texts? Broadly, a narrative is a text that tells a story. Although often treated as synonyms, narrative and story are distinct. A story is an event or sequence of events, and a narrative is a story plus

narrative discourse, where narrative discourse is the way events are represented (Abbott, 2008). In other words, a narrative is a specific material representation of a story—it is the telling of a story.

Distinguishing between narratives and non-narrative texts is not always easy and scholars disagree on what features should be used to do so (Abbott, 2008). One approach is to distinguish narratives from non-narratives based on elements of the text according to a structural definition. Structural definitions consider a narrative to be any text which contains all the necessary elements of narrative. What those elements are is the subject of scholarly debate. Abbot (2008) provides a “bare minimum” definition of narrative as, “the representation of an event or a series of events” (p. 13). But as Abbot acknowledges, even that most basic of definitions is contested. It is inclusive enough that all narratives represent an event or series of events, but different scholars provide more stringent requirements than this for a text to qualify as a narrative. For example, elements such as conflict, setting, character, chronology, and theme are sometimes deemed necessary (Abbot, 2008). Additionally, narrative is often reserved for specific kinds of representations, namely written or spoken language (Rimmon-Kenan, 2003).

Instead of focusing on the features of the text itself, cognitive definitions of narrative are concerned with how texts are received or meant to be received by their readers (Oatley, 2017; Ryan, 2005). That is to say, they distinguish narrative and non-narrative texts not based on the content of the text itself but on whether that text functions as a narrative. As with structural definitions, there are different ways of defining whether a text functions as a narrative. For Oatley, functioning as a narrative

means readers engage with the text via mental simulation of a social world (Oatley, 2017).

Ryan defines narrative more precisely as any text intended to inspire a narrative response, here referring to a specific kind of mental representation which includes imagining: (a) a world with characters and objects, (b) events which cause changes to the status quo, and (c) causal and other associations between events that give events coherence and form them into a plot (Ryan, 2005).

What constitutes a non-narrative depends on what definition of narrative is used. If a structural definition is used, a non-narrative is any text that is missing one or more essential narrative elements according to the definition used. For example, this paper is not a narrative as it does not represent an event or sequence of events—it lacks plot. Argumentative essays and opinion pieces represent ideas, opinions, or arguments; although they may use narrative within them, they are not narratives because they do not represent events. News reports often do represent events but are not typically narratives because they lack other important elements. For example, a news report might recount the events that occurred but not include details of the setting, may not centre around story characters, and the events may not unfold sequentially. As such, a news report might recount events without representing them and may not be intended to inspire a narrative response.

Episodic Memory

What Ryan (2005) calls a narrative response and Oatley (2017) calls simulation may both be instantiations of the same processes involved in episodic memory. Tulving (1972) proposed episodic memory to distinguish between semantic memories

(knowledge of words and their meanings, concepts, and classifications) and episodic memories (memories of personal experiences and their temporal relations).

Remembering one's past involves constructive episodic simulation—an active process whereby stored information about a past experience is combined with schematic information to simulate the experience of a sequence of events (Addis, 2018).

Subjectively, this kind of remembering feels like being transported to a different time and place and re-experiencing the events of the memory (Zaman et al., 2024). This includes external perceptions—what one sees, hears, smells, and feels—and internal perceptions—the thoughts, emotions, and bodily sensations of the experience (Addis, 2018). Despite being entirely the product of imagination, these perceptions can feel very real.

Although Tulving (1972) originally proposed episodic memory to describe personal memories of past experiences, the same process—constructive episodic simulation—is also used to imagine novel episodes. This includes internally-generated episodes, as well as simulating narrative worlds (Addis, 2018; Schacter et al., 2007). Readers who are transported (i.e., immersed, absorbed, engrossed) experience and encode memories of narratives as episodic memories—as if they were themselves limited participants in the story (Gerrig, 2023; Gerrig & Jacobina, 2009). This includes the subjective sense of being transported away from reality and into the time and place of the narrative, and the feeling of experiencing—or at least witnessing—the events themselves (Busselle & Bilandzic, 2009).

Narrative and Emotion

Stories are emotional (Busselle & Bilandzic, 2009; Van Laer et al., 2014). Engaged readers report frequent and intense emotion shifts, especially during moments of heightened suspense or tension for story characters (Schmidt et al., 2023; Winkler et al., 2023). Narratives may affect emotions in a variety of ways, including in ways that are unlikely to elicit empathic concern, such as responding to the aesthetic beauty of the prose, or evoking memories of one's own emotional memories (for complete taxonomies of narrative emotions, see Habermas, 2018; Oatley, 1995). Narratives also elicit emotions through simulation of the narrative's events. As readers simulate the experience of the narrative, it is as if they observe and are present with story characters in their journey. This explains why emotion shifts are strongest during moments when the stakes are highest for characters in the story. Readers' emotions reflect their concern for the wellbeing of story characters. They feel this way because, even if they are aware that characters and events are fictional, the characters and events feel real.

Eliciting Empathic Concern

When narratives tell the story of someone in need, they are likely to elicit empathic concern. Empathic concern refers to other-oriented emotions felt for a person seen as in need and congruent with a desire for that person's welfare (Batson, 2023a). It includes many different subjective feelings, including sympathy, tenderness, warmth, and compassion (Batson, 2023a). Importantly, these feelings are felt for, and not necessarily with a person seen as in need. A person experiencing empathic concern is not vicariously feeling the emotions of someone else, they are feeling their own feelings for what they see as a discrepancy between the way things are and the way they should be

for another person to experience complete wellbeing. That is, empathic concern is felt out of concern for someone else's welfare.

People feel empathic concern when they perceive of another person as in need. An important feature of empathic concern is that it is grounded in the perceiver's perspective (Batson, 2023c). The perspective of the individual who is the target of empathic concern is inconsequential. If a person does not see a need in another who sees themselves as in need, that person will not experience empathic concern. Conversely, if a person sees a need in an individual who is entirely satisfied with their circumstances, that person may feel empathic concern anyway. This means that empathic concern is not so much elicited by observing a person in an aversive or undesirable situation, but by observing how a person responds to that situation. People will feel empathic concern for a person who receives an electric shock or a needle injection only if that person responds as if they are in pain (Berger, 1962; Lamm et al., 2010). People will also feel empathic concern for someone who experiences what would not normally be a painful stimulus if they believe it causes the person harm (Lamm et al., 2010).

These studies suggest that at least two pieces of factual information are necessary for eliciting empathic concern: an aversive stimulus, and an aversive response. Such information may be communicated via narrative or non-narrative messages. Narratives may be more effective than non-narratives at eliciting feelings of empathic concern. Narratives are often deeply emotionally impactful. Readers who are deeply engaged in a narrative report frequent and intense shifts in their emotions (Schmidt et al., 2023; Winkler et al., 2023). One reason for this may be that narratives feel present (Lee, 2004).

The engaged reader feels present and close to the narrative world, the people and events feel real and important, and their emotions respond accordingly. For this reason, narrative accounts may be more emotionally affecting than non-narrative depictions of the same or similar circumstances (Hamby et al., 2018). When a reader is engaged in the narrative, and simulates the experience of the narrative events, they remember those events as if they were there and witnessed them happen, the memories of the narrative events are encoded as episodic memories, and it feels and is remembered as a personal experience (Gerrig, 2023; Gerrig & Jacobina, 2009). This may give the circumstance conveyed in the narrative emotional weight. By comparison, the same information being conveyed in a non-narrative way—delivering factual information about an individual or group and their circumstances without a narrative structure—tells the reader that a person is in need but may be less likely to cause the reader to imagine the experience. They may therefore feel less close to and less emotionally impacted by the knowledge.

The Current Study

Observing a person in a challenging situation and seeing that situation having a detrimental effect on their wellbeing is often sufficient to elicit feelings of empathic concern. Such information can be conveyed via narrative by telling the story of an individual struggling with a challenging experience. It can also be conveyed without narrativity, such as by delivering factual information. These two modes of conveying information about individuals or groups in need may have different effects on readers. I hypothesized that readers would feel greater empathic concern after reading narrative

compared to non-narrative messages. I tested this hypothesis by conducting a systematic review and meta-analysis.

Method

Search and Screening

All steps of this systematic review and meta-analysis were conducted by a single reviewer. Potentially eligible sources were identified using the EBSCO interface to search the databases PsycInfo, PsycArticles, Business Source Ultimate, and Communications & Mass Media Complete using the search terms: narrative AND empath* AND (experiment or trial).

Sources identified with this search were first screened by title and abstract only. Sources that could not be excluded based information in the title and abstract alone proceeded to full-text screening. Studies which met the eligibility criteria were included in the final analysis.

Other potentially relevant sources were identified by searching the references of eligible studies. If the title of a referenced source suggested the source might be relevant, the source was added to the title and abstract review.

Eligibility

A source was eligible for inclusion in the present review if:

1. it was an English-language publication in a peer-reviewed academic journal,
2. the sampling population was adults aged 18 or older,
3. participants were randomly assigned to experimental conditions,
4. one or more experimental conditions exposed participants to a text-based narrative message,

5. one or more experimental conditions exposed participants to a text-based non-narrative message,
6. the narrative and non-narrative messages conveyed a message about an individual or group in need, and
7. participants reported empathic concern after the intervention.

Extraction

The primary data extracted from eligible studies were the sample sizes, means, and standard deviations on the measure of empathic concern in the narrative and non-narrative experimental conditions. Other information extracted included the word counts of narrative and non-narrative messages, the scale used to measure empathic concern, and standard details about the study (date of data collection, funding sources, etc.). The complete extracted data are shared in a public repository at

https://www.notion.so/226b56242af3806c8ea6e7dd8cf06743?v=227b56242af38057a047000c572aaba&source=copy_link and https://www.notion.so/227b56242af38007808bcca0ab54e2ed?v=227b56242af3809ebf75000cf37d25aa&source=copy_link.

Analytic Strategy

Analyses were conducted in R (R Core Team, 2024) using the *metafor* package (Viechtbauer, 2010).

Bias-corrected standardized mean differences (i.e., Hedge's *g* values) between conditions were calculated for each study (Viechtbauer, 2010). The main statistical model was a multivariate meta-analytic model with a random effect of study. The random effect was included to account for dependence caused by multiple effects coming from the

same source. This occurred because some articles reported on multiple studies and because some study designs included several narrative or non-narrative conditions. Significance was inferred using the traditional alpha of .05.

Standardized residuals and Cook's distances were calculated and visualized to look for outliers and influential cases; however, no plans were made to omit such cases from analysis. A funnel plot was generated to visualize the possible presence of publication bias.

Results

The initial search yielded 133 sources for screening. An additional 6 sources were added to screening from the citation search. After excluding irrelevant articles based on title and abstract screening ($n = 110$), the remaining ($n = 19$) advanced to full-text screening. From those, 5 studies were deemed eligible for inclusion, including two studies from the same source (i.e., the published article reported the results of two studies, both of which were eligible for inclusion). Reasons for exclusion were not having a non-narrative message condition ($n = 9$), not being a peer-reviewed journal article ($n = 4$), and not measuring empathic concern ($n = 2$).

Features of the eligible studies are presented in Table 1. Studies measured empathic concern using the Interpersonal Reactivity Index (IRI; Davis, 1980) or Batson's empathic concern adjectives (Batson et al., 1997). The IRI is a multidimensional measure of empathy and includes an empathic concern subscale. It is meant to measure dispositional empathic concern as a stable personality trait—the extent to which a person is likely to feel empathic concern when observing another in need. The empathic concern

items of the IRI are phrased to ask about a person's disposition but are otherwise similar to Batson and colleagues' measure (Batson et al., 1997). Batson et al. (1997) measured state empathic concern by asking participants to rate how much each of six different adjectives describe their current feelings: *sympathetic*, *softhearted*, *warm*, *compassionate*, *tender*, and *moved*. Johnson (2013) used all six items and Oliver et al. (2012) used only *sympathetic*, and *compassionate*.

The main meta-analysis model revealed a statistically significant effect of narrative on empathic concern, $d = 0.56$, $SE = 0.27$, $z = 2.09$, $p = .036$, 95% CI [0.036, 1.09]. Cochran's Q test (Cochran, 1954) for heterogeneity was also significant, $Q(4) = 55.5$, $p < .001$, suggesting that there is greater heterogeneity in effect sizes than expected based on sampling variability alone. A forest plot of effect sizes for the five studies is shown in Figure 1.

A second meta-analysis model with measurement instrument included as a moderator was also fit. There was a significant moderating effect of measurement instrument, such that the effect of narrative on empathic concern was smaller for studies using the IRI ($d = 0.02$) than for studies measuring state empathic concern ($d = 0.90$), $d = -0.88$, $SE = 0.28$, $z = -3.10$, $p = .002$, 95% CI [-1.44, -0.33]. Cochran's Q test was significant for this model, $Q(3) = 9.97$, $p = .02$, suggesting that the residual heterogeneity in effect sizes was larger than expected due to sampling variability even after accounting for the moderating effect of measurement instrument.

Publication bias was assessed by visual inspection of a funnel plot (see Figure 2). Asymmetrical plots can be an indicator of publication bias (Light & Pillemer, 1984). There

is some asymmetry in Figure 2. The two experiments from Johnson et al. (2013) have larger effect sizes and smaller samples than the other studies. There are no studies with small effects and small sample sizes, which is consistent with unpublished null results. That is, if there were a file-drawer problem wherein studies that would be eligible for inclusion in this review were left unpublished because they had null results, those studies would likely be in the bottom left of the plot. The fail-safe N for this model was 2, meaning that at least two such unpublished file-drawer studies with null results would need to be added to the model to reduce the significance level of the model below the alpha level of .05. In summary, results suggest that a file-drawer problem is plausible and could change conclusions; however, it is also plausible that there is no file-drawer problem and there just are not many experimental studies of the effect of narrative on empathic concern.

Discussion

This review examined the effect of narrative on eliciting empathic concern. The primary hypothesis was that narrative messages would cause readers to experience more empathic concern than non-narrative messages. This hypothesis was supported. There was a significant overall causal effect of narrative on empathic concern. This effect was qualified by a significant interaction. A single narrative message did not cause changes in dispositional empathic concern as measured by the IRI; one short narrative message was sufficient to cause substantially more state empathic concern compared to a corresponding non-narrative message.

The three studies reviewed that measured state empathic concern all showed considerably more empathic concern in the narrative than the non-narrative conditions. But was this difference caused by narrativity? In the experiments conducted by Johnson and colleagues (2013), the non-narrative condition still included the main plot elements. They differentiated narrative from non-narrative based on the presence of descriptive language and internal dialogue. The non-narrative message was also considerably shorter (491 words compared to 3,108 in the narrative condition). The increased empathic concern in the narrative condition could be attributed to the descriptive language of the narrative text more deeply engaging readers in the story. It could also be attributed to the narrative condition including the protagonist's internal monologue. In this case, the effect would be less a result of narrativity and more to do with the narrative including different information that changed readers' perceptions or valuing of the protagonist.

Evidence of narrative causing empathic concern is stronger for the other study of state empathic concern. In Oliver and colleagues' (2012) design, the narrative and non-narrative conditions differed considerably with respect to many structural narrative elements. The narrative articles were rich with all the basic elements of narrative. They followed the story of a main protagonist, detailing a sequence of events related to struggles with obtaining healthcare using vivid imagery. The non-narrative articles shared much of the same information, including some of the plot elements of the corresponding narrative article but the arguments were not framed around the narrative. They were presented later in the article and were described briefly as a means of supporting the factual arguments made previously. The narrative and non-narrative

articles were also closely matched with respect to word count. This suggests that it was the narrative elements and not some other differences that drove the differences in empathic concern between conditions in this study.

Taken together, findings do suggest that narratives elicit empathic concern more than non-narrative texts. There is a causal relationship between narrative and empathic concern. Nevertheless, this conclusion is based on only three experiments from two published articles. There is a need for future research to increase confidence in this conclusion and to shed further light on the magnitude of the effect and the conditions necessary for it to be observed.

Fiction and Empathy

A related area of research is that showing an effect of reading fiction on empathy. Scholars argue that reading fiction increases empathy across the lifespan (Bal & Veltkamp, 2013; Johnson et al., 2013; Mar & Oatley, 2008). One proposed explanation for why reading fiction increases empathy is that reading increases sense imagery vividness and stronger mental imagery increases empathy for others (Johnson et al., 2013). Evidence for this comes from an experimental study showing that imagery training prior to reading increases empathic concern and helping behaviour (Johnson et al., 2013). The current research suggests that reading a single narrative does not increase dispositional empathy. Rather, the immediate effect of narrative is on state empathy for story characters and those like them and growth in capacity to feel empathy occurs with narrative engagement over prolonged periods of time. Nevertheless, it is also difficult to link the present findings to the literature showing an effect of reading on empathy

because of the ambiguity of the term empathy. The aforementioned study did measure empathic concern specifically; however, empathy is used to refer to many distinct but related phenomena (Batson, 2009; Hall & Schwartz, 2019).

Limitations and Future Directions

One obvious limitation of this review is the relatively small number of studies. Only five eligible studies (from 4 articles) were identified. Of the included studies, two measured trait empathic concern with the IRI and three measured state empathic concern. Given that the effect of interest was for state empathic concern, this means that conclusions are based on results of just three studies—two of which came from the same article. The small number of eligible studies still yielded a large enough sample to detect small effects (80% power for $d = 0.15$, 95% power for $d = 0.19$, and 99% power for $d = 0.23$); however, a larger pool of eligible studies would increase confidence that the effects were not driven by idiosyncrasies of a particular design, publication bias, or some other confounding factor. This may be especially important because two of the three studies had small samples ($n = 29$ and $n = 38$ per group, respectively). While individual studies having small samples is not a problem in meta-analysis *per se*, it may present an issue here because more than half the observations came from a single study. Part of the value of meta-analyses is that combining effect sizes across multiple studies limits the influence of individual studies which may be outliers due to systematic or random variance. For the present meta-analysis in which a single study represents more than half of observations, a single study being a poor representation of the true effect could have substantial impact on the magnitude of the effect or even change conclusions altogether.

There were too few studies to adequately assess publication bias. The funnel plot (Figure 2) shows an absence of effects on the bottom left—studies with small samples and small effects. This is consistent with what would be expected from publication bias because studies with small samples and null findings are the least likely to be published. But given the small number of overall studies, the asymmetry in the funnel plot may also be explained by few studies having been conducted on the effect of narrative on empathic concern. If the asymmetry is due to publication bias, then finding and including unpublished studies could change the magnitude of the observed effect and the conclusions of the significance test. The fail-safe n calculation indicated that as few as two unpublished studies with null findings could be enough to increase the overall significance level above .05, therefore changing the conclusions of the meta-analysis.

I proposed that the relationship between narrative and empathic concern was due to narrative causing readers to have a narrative response (i.e., engage in constructive episodic simulation), which would in turn cause them to perceive a greater need (when a need is conveyed by the narrative) and place greater value on the character's welfare. These two factors are proposed necessary preconditions for empathic concern and would therefore lead to greater empathic concern. The current study is consistent with this model; however, it does not directly support any of the individual causal pathways. Rather, it demonstrates a causal relationship between narrative and empathic concern, which is a downstream effect of narrative in my proposed causal model. Other research does support some of these other causal pathways. For example, there is causal evidence for an effect of perspective taking on empathic concern (Batson et al., 1997; Batson,

2023). Nevertheless, further research is needed to support or refute the complete causal chain. Note that cross-sectional mediation analyses are not normally sufficient to provide causal evidence. Testing causal mediation is challenging and typically requires accumulating evidence across multiple studies (Bullock et al., 2010).

For the current study, I did not provide a specific definition of narrative. This was for several reasons. One was that the primary aim of the study was to understand the differential effects of narrative and non-narrative texts as defined by a cognitive definition of narrative. That is, the effect of interest was comparing texts that inspire a narrative response and texts conveying the same or similar information that do not inspire a narrative response. Structural definitions that are divorced from this cognitive definition are problematic for this purpose because they may inadvertently categorize texts that inspire narrative responses as non-narratives and *vice versa*. This could be especially important for strict definitions of narrative that require multiple elements. If a text contains several elements of narrative—plot, character, conflict—but is missing or lacking in another, does it no longer qualify? How much of each element is necessary? These are important questions that should be addressed in studies comparing narrative and non-narrative texts; however, as the current study was a review of extant literature, providing a specific definition risked conflicting with the definitions used by other researchers and therefore limiting the eligible studies. As such, I used the definitions of narrative and non-narrative employed by authors of studies in the review.

There are many definitions of narrative in the literature. Additionally, narrative is often defined according to a collection of several elements that are necessary to

constitute a narrative text (e.g., character, plot, conflict, setting). These can each be manipulated independently. In other words, there are many ways for a narrative to not be a narrative. A non-narrative text could be non-narrative because it lacks any one of the necessary elements needed to constitute a narrative, but that does not mean that each element of a narrative is equally important for eliciting empathic concern. An interesting avenue for future research would be to investigate differences in empathic concern caused by emphasizing or deemphasizing specific narrative elements. That is, instead of examining the effect of narrative on empathic concern, examining the respective effects of aspects of character, plot, conflict, and setting on empathic concern.

Conclusion

This review and meta-analysis showed that there is an effect of narrative on eliciting empathic concern for story characters. Nevertheless, as this effect was demonstrated primarily based on the findings of three experiments from two articles, there is still room for further replications. There is also a need for more precise operational definitions of narrative and non-narrative in order to establish the boundary conditions of this effect.

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Tables

Table 1.

Features of Included Studies

Study	$n_{\text{non-narrative}}$	$n_{\text{narrative}}$	Summary of Non-narrative	Summary of Narrative	Measure	SMD
Andrews et al., 2022	319	317	“Five messages that conveyed the same information as the narrative messages but were written in an expository format. ... They were adapted from the narratives to present a matter-of-fact account of the same information from a third-party perspective without a central character.”	“Five first-hand experiential narratives from the perspective of incarcerated people ($n = 160$) or people working in healthcare ($n = 157$) during the COVID-19 pandemic.”	IRI	-0.07
Walkington et al., 2020	117	118	“Newspaper article version of the same information as was presented in the narrative condition	“Two chapters from ‘Rachel’s Holiday’ by Marian Keyes (1998). This novel is about a young woman who	IRI	0.12

Study	$n_{\text{non-narrative}}$	$n_{\text{narrative}}$	Summary of Non-narrative	Summary of Narrative	Measure	SMD
			(e.g., information about a drug user) ... presented to participants as a factual newspaper article” (p. 4).	experiences difficulties in her life relating to drug use and associated behaviours (such as stealing and deceiving friends)” (p. 4).		
Oliver et al., 2012	199	199	A news story about a policy presentation with factual information about a healthcare issue. Manipulated whether the issue was faced by elderly persons, immigrants, or prisoners.	A narrative news story about the same healthcare issue as in the non-narrative condition presenting the same factual information through the story of a specific person. Also three different stories, one for an elderly person, one for an immigrant, and one for a prisoner.	EC	0.62
Johnson, 2013, Study 1	29	29	"the same counterstereotypical exemplars [as the full narrative] but without descriptive language,	Excerpt from “Shaila Abdullah’s fictional novel, Saffron Dreams (2009, pp. 57–64). The protagonist is an	EC	0.70

Study	$n_{\text{non-narrative}}$	$n_{\text{narrative}}$	Summary of Non-narrative	Summary of Narrative	Measure	SMD
			dialogue, or monologue.”	educated and strong-willed Muslim woman who is assaulted in a New York City subway station. She is also pregnant and exhibits extraordinary courage by standing up to her attackers’ ethnic and religious slurs.”		
Johnson, 2013, Study 2	38	38	Same as Study 1	Same as Study 1	EC	0.55

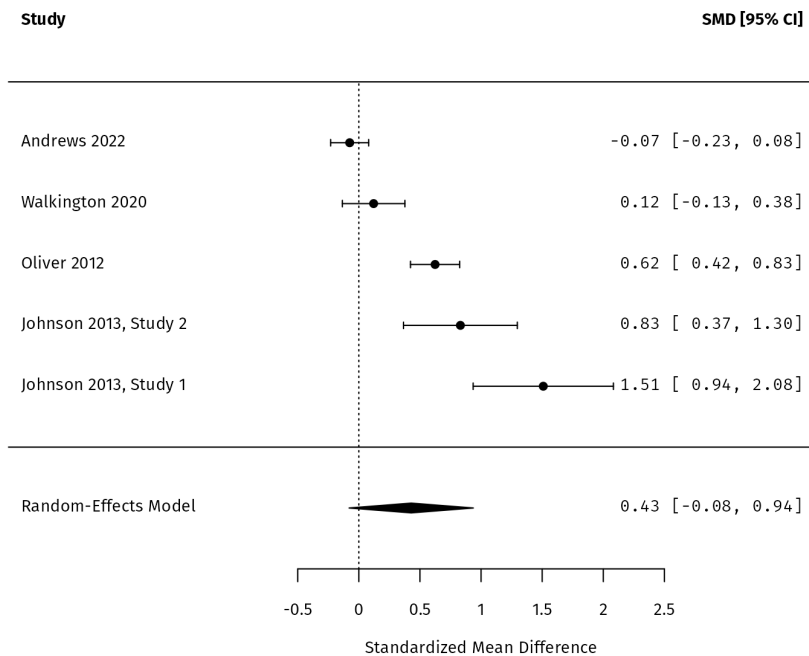
Note. EC = empathic concern measured using some or all of the items from Batson et al. (1997); IRI = Interpersonal

Reactivity Index (Davis, 1980). SMD = standardized mean difference.

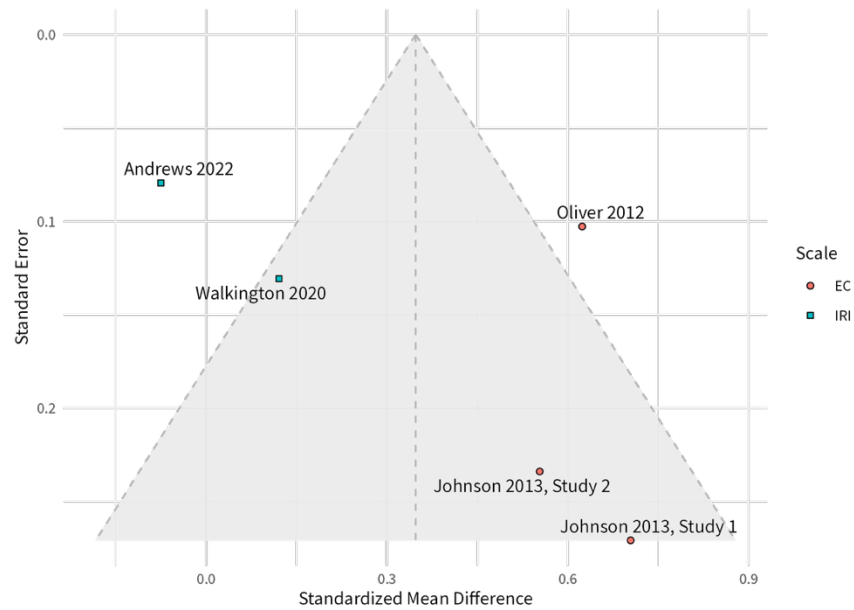
Figures

Figure 1.

Forest Plot



Note. SMD = standardized mean difference. The two studies with non-significant effects near zero measured empathic concern as a disposition or tendency to be moved by a person in need. The three studies with significant positive effects measured empathic concern as a transient emotional state.

Figure 2.*Funnel Plot.*

Note. EC = empathic concern measured using some or all of the items from Batson et al. (1997). IRI = Interpersonal Reactivity Index (Davis, 1980). The main distinction between the IRI and Batson et al.'s (1997) measure is the IRI treats empathic concern as a stable personality trait and the EC measures empathic concern as a transient emotional state.